

Lorenzo Corti

Residence: Netherlands – Nationality: Italian

🏠 <https://lcorti.github.io> — ✉ corti.lorenz@gmail.com — 💼 [linkedin.com/in/lorenzo-corti-46a22b181](https://www.linkedin.com/in/lorenzo-corti-46a22b181) — 🌐 github.com/LCorti

Summary

I research and develop **methods** and **tools** to diagnose large vision and language models through a human-centred approach. My work bridges **Natural Language Processing**, model **Interpretability**, and **Human-Computer Interaction** to enable the sociotechnical diagnosis of vision and language models. Specifically, I frequently rely on frameworks like Hugging Face **Transformers**, **spaCy**, and **scikit-learn** to self-host and run experiments on open-source vision and language models. I am currently pursuing a PhD at Delft University of Technology. I worked as a research assistant at Politecnico di Milano, where I contributed to two EU H2020 projects (TRIGGER and PERISCOPE) after receiving my MSc and BSc in Computer Science and Engineering from the same institution.

Skills

Languages	Python, Java, C, Javascript	Machine Learning	NumPy, Pandas, scikit-learn, spaCy
Data Management	Relational DBs and NoSQL	Deep Learning	Pytorch, Captum, Huggingface
Web Development	Spring, FastAPI, Django		Transformers, Datasets, and Evaluate
DevOps	Docker, VPS, Git	Project Management	Research design, planning, and execution

Education

Delft University of Technology, Delft, NL **Feb 2022 – 2027***
PhD Candidate in Computer Science **Finalising my dissertation and can terminate earlier than Feb. 2027.*

- *Dissertation:* Beyond Benchmarks – Sociotechnical Diagnosis of Vision and Language Model Behaviours.
- *Supervision:* Prof.Dr.Ir. Geert-Jan Houben (promotor) and Dr. Jie Yang (copromotor).

Politecnico di Milano, Milan, IT
Master of Science in Computer Science and Engineering (110/110) *Sept 2017 – Dec 2019*

- *Thesis:* Time Conditional Generative Adversarial Networks for Augmentation of Irregularly Sampled Time Series.
- *Supervision:* Marco Brambilla (Politecnico di Milano) and Pavlos Protopapas (Harvard University).

Bachelor of Science in Computer Science and Engineering (98/110) *Sept 2014 – July 2017*

- *Thesis Project:* Development of a customer relationship management system for an advertising company.
- *Deliverables:* Requirement analysis with i* framework, UML design, and documented codebase.

Professional Experience

Doctoral Candidate **Feb 2022 – 2027***
Delft University of Technology, Delft, Netherlands

- **End-to-End Diagnosis of Vision Language Models:** Method to diagnose the behaviours of VLMs, verify observable ones, and quantify concept-level effects. Evaluated on four VLMs across four benchmarks for multiple-choice and open-ended visual question answering tasks. Concept-level effects are quantified using a Double-ML causal inference estimator, fit onto synthetically occluded visual inputs. *Under Review.*
- **Toolkit for LLM Behaviour Diagnosis:** Design and development of a toolkit to diagnose LLM behaviours on known linguistic structures. The toolkit supports encoder-only, encoder-decoder, and decoder-only transformer-based LLMs, different NLP datasets, known NLP metrics, inspection and tagging of LLM outputs, and behavioural resting and reporting. The tool is validated through a qualitative inquiry with NLP industry practitioners. *Under Review.*
- **Situating Goodness Criteria for AI Behaviours:** Qualitative inquiry into the relationship between situational factors and the goodness criteria for AI behaviours. The work captures clinicians' informational needs and their evolution over time for the behaviours of a hypothetical AI-based clinical decision support system. *Published at CHI 2024.*
- **Extraction of Behaviours of Vision Models:** Explainable AI method to extract observable behaviours of CNN-based vision models. Applied the method to vision models in the presence of known biases (introduced via fine-tuning) and evaluated in terms of fidelity, coherence, and accuracy of the behaviours extracted. *Published at HCOMP 2022.*

Research Assistant **Feb 2020 – Jan 2022**
Politecnico di Milano, Milan, Italy

- I was part of two EU H2020 projects, TRIGGER and PERISCOPE. The COCTEAU and PERSEUS platforms (below) were both first designed and implemented within the scope of the TRIGGER project, and later extended to support the goals and tasks of the PERISCOPE project.
- **COCTEAU:** Gamified, crowdsourcing tool that has the goal of enabling large-scale citizen engagement and co-creation in support of policymakers when approaching complex governance solutions.

- **PERSEUS**: Integrated toolkit within the TRIGGER project linked to the concept of actorness and supported by data science techniques like named entity recognition, topic extraction, sentiment analysis and language models. PERSEUS also includes a document and resource search functionality powered by the Elastic stack.
- **IDiOM**: IDiOM is a NLP data analysis pipeline that collects data from social and news media sources. It implements data collection from social and news media sources and runs natural language processing and machine learning analyses. The final purpose is to create a publicly available dashboard raising citizen awareness about misinformation.
- **VaccinItaly**: Analysis of the Italian discourse on COVID-19 (perception and vaccine intake/hesitancy) on social media (Twitter and Facebook) and news outlets. Includes temporal, geographical and lexical analyses as well as training of stance classifiers (pro- or anti-vaccination).

Mentoring & Teaching Experience

Mentoring Experience

Topics: NLP, language models, model interpretability, human computation, and information systems.

- 8 M.Sc. theses
- 5 B.Sc. theses
- 10 student group projects (4-6 students each)

Teaching Experience

Fundamentals of Artificial Intelligence Programme, TU Delft

2022 – 2024

- Creation of teaching materials, support on-campus sessions, and grading assignments for the following topics: data management systems, data work, interpretable & explainable AI, and natural language processing.

Crowd Computing, TU Delft

2022 – 2024

- Grading students' reviews of influential works in human computation and mentoring group projects.

Post-graduate Master Courses, Quantia Consulting and Cefriel

2021 – 2022

- Cloud Data Engineering: basics of Information Retrieval and practical sessions with NoSQL technologies (MongoDB, Neo4j, Cassandra, and the Elastic Stack). Offered to BIP Consulting.
- Advanced Data Science: theory and practice on Generative Adversarial Networks. Offered to Allianz and Nestlé.

Computer Science Fundamentals, Politecnico di Milano

2021

- Supporting exercise and practical sessions, mentoring group projects, and grading assignments.

Software Methodologies and Architectures for Security - Enterprise ICT Architectures, Politecnico di Milano

2020 – 2021

- Supporting exercise and practical sessions, mentoring group projects, and grading assignments.

Event Organisation

1. The 25th International Conference on Web Engineering (ICWE 2025): Proceedings Co-chair.
2. The 12th AAAI Conference on Human Computation and Crowdsourcing (HCOMP 2024): Web Co-chair.

Languages

- **Italian**, Native.
- **English**, Full Professional Proficiency (B2 FCE, Cambridge).
- **Spanish**, Elementary (A2).

Publications

1. *End-to-End Diagnosis of Vision Language Models*. **Lorenzo Corti**, Jie Yang. *Under review*.
2. *Toolkit for LLM Behaviour Diagnosis*. **Lorenzo Corti**, Sebastiaan Beekman, Jie Yang. *Under review*.
3. *A.I. Robustness: a Human-Centered Perspective on Technological Challenges and Opportunities*. Andrea Tocchetti*, **Lorenzo Corti***, Agathe Balayn*, Mireia Yurrita, Philip Lippmann, Marco Brambilla, Jie Yang (* equal contribution). In: ACM Computing Surveys. 2025. DOI: <https://doi.org/10.1145/3665926>.
4. *Desirability vs. Feasibility: A Research through Design Inquiry of Explainable AI*. **Lorenzo Corti**, Jie Yang. In: Bi-directionality in Human-AI Collaborative Systems. 2025. DOI: <https://doi.org/10.1016/B978-0-44-340553-2.00010-1>.
5. *It Is a Moving Process: Understanding the Evolution of Explainability Needs of Clinicians in Pulmonary Medicine*. **Lorenzo Corti**, Rembrandt Oltmans, Jiwon Jung, Agathe Balayn, Marlies Wijsenbeek, Jie Yang. In: Proceedings of the CHI Conference on Human Factors in Computing Systems. 2024. DOI: <https://doi.org/10.1145/3613904.3642551>.

6. *CHIME: Causal Human-in-the-Loop Model Explanations*. Shreyan Biswas, **Lorenzo Corti**, Stefan Buijsman, Jie Yang. In: Proceedings of the AAAI Conference on Human Computation and Crowdsourcing. 2022. DOI: <https://doi.org/10.1609/hcomp.v10i1.21985>.
7. *EXP-Crowd: A Gamified Crowdsourcing Framework for Explainability*. Andrea Tocchetti, **Lorenzo Corti**, Marco Brambilla, Irene Celino. In: Frontiers in Artificial Intelligence. 2022. DOI: <https://doi.org/10.3389/frai.2022.826499>.
8. *Scaling collaborative policymaking: how to leverage on digital co-creation to engage citizens*. Andrea Tocchetti, Diletta Di Marco, **Lorenzo Corti**, Marco Brambilla. In: Data for Policy. 2021. DOI: <https://doi.org/10.5281/zenodo.5227880>.
9. *A Web-Based Co-Creation and User Engagement Method and Platform*. Andrea Tocchetti, **Lorenzo Corti**, Marco Brambilla, Diletta Di Marco. In: Proceedings of International Conference on Web Engineering. 2021. DOI: https://doi.org/10.1007/978-3-030-74296-6_38.

Workshop and Demo Papers

1. *Understanding Stakeholders' Perceptions and Needs Across the LLM Supply Chain*. Agathe Balayn, **Lorenzo Corti**, Fanny Rancourt, Fabio Casati, Ujwal Gadiraju. In: ACM CHI Workshop on Human-Centered Explainable AI. 2024. DOI: <https://doi.org/10.48550/arXiv.2405.16311>.
2. *ARTIST: ARTificial Intelligence for Simplified Text*. **Lorenzo Corti**, Jie Yang. In: Generative AI and HCI workshop at CHI. 2023. DOI: <https://doi.org/10.48550/arXiv.2308.13458>.
3. *COCTEAU: an Empathy-Based Tool for Decision-Making*. Andrea Mauri, Andrea Tocchetti, **Lorenzo Corti**, Yen-Chia Hsu, Himanshu Verma, Marco Brambilla. In: Companion Proceedings of the Web Conference. 2022. DOI: <https://doi.org/10.1145/3487553.3524233>.
4. *VaccinItaly: monitoring Italian conversations around vaccines on Twitter and Facebook*. Francesco Pierri, Andrea Tocchetti, **Lorenzo Corti**, Marco Di Giovanni, Silvio Pavanetto, Marco Brambilla, Stefano Ceri. In: Workshop Proceedings of the 15th International AAAI Conference on Web and Social Media. 2021. DOI: <https://doi.org/10.36190/2021.11>.
5. *A content-based approach for the analysis and classification of vaccine-related stances on Twitter: the Italian scenario*. Marco Di Giovanni, **Lorenzo Corti**, Silvio Pavanetto, Francesco Pierri, Andrea Tocchetti, Marco Brambilla. In: Workshop Proceedings of the 15th International AAAI Conference on Web and Social Media. 2021. DOI: <https://doi.org/10.36190/2021.52>.